Approach to the quantification of hanging-wall behaviour

D. M. Milne, R. C. Pakalnis and P. J. Lunder

Synopsis

During the mining of an open stope deformation of the hanging-wall and the potential for dilution generally increase with the size of the exposed hanging-wall. Instrumentation and survey data are often available and give localized measurements of hanging-wall behaviour. However, it is difficult to project these local measurements to obtain an overall view of deformation or dilution and it is also not easy to predict the behaviour of the hanging-wall with continued mining.

Without considering discrete structures that may delineate unstable blocks, the mechanisms that drive hanging-wall deformation and possible dilution comprise several factors, which can include elastic, plastic and discontinuum deformation. These stages of deformation are not mutually exclusive and at any point during mining all three of these mechanisms may be acting on the hanging-wall rock mass. For a given rock mass and stress regime the magnitude of deformation generally increases as the distance to supporting abutments increases. Computer modelling techniques are available to analyse the three stages of deformation mentioned, but the difficulty lies in trying to obtain realistic rock mass properties for input and in determining when each stage of deformation is acting.

As surface deformation increases it can generally be assumed that the potential for surface dilution also increases. Instrumentation data on hanging-wall deformation and dilution can be used to predict the effect of continued mining; however, it is often difficult to calibrate numerical models against field data owing to the large number of input parameters required. An empirical approach is recommended that uses two new parameters termed the 'radius factor' (RF) and the 'effective radius factor' (ERF). These terms are based solely on the distance from a point on a surface to the supporting abutments. A method is proposed for predicting the effect of continued mining from existing deformation and dilution data for a given hanging-wall. Case histories are presented that look at both hanging-wall dilution and deformation and relate them to the ERF term.

Hanging-wall geometry and behaviour

In Canadian open-stope mining very large open hanging-walls, of the order of 100 m along strike and dip, can be created. As mining progresses a zone of low stress develops where the stresses normal to the hanging-wall are shed to the abutments. This zone of elastic relaxation increases in depth as the distance to abutments increases and has been depicted graphically by Hoek and Brown.[1] As the magnitude of elastic deformation increases the ground response may change from an elastic to a plastic deformation. The mechanism that drives hanging-wall deformation can generally be divided into three stages:

(1) With initial exposure elastic relaxation and the redistribution of stress around the hanging-wall contribute to most of the movement.

(2) With increased exposure the effective elastic modulus of the rock mass may start to decrease as fractures in the rock mass begin to dilate.

(3) If hanging-wall exposure continues to increase, the rock mass around the immediate open surface may cease to act as a continuum and an arching or discontinuum mechanism may control the deformation.

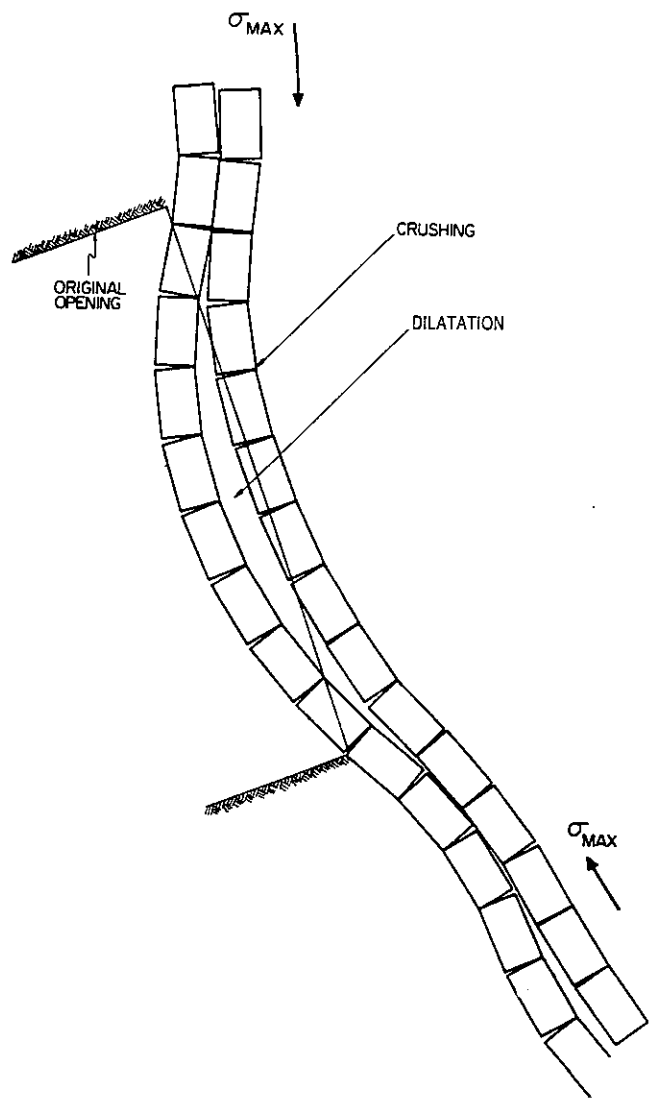


Fig. 1 Schematic diagram of stope hanging-wall showing general *voussoir*-arch deformation

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In most Canadian open-stope mines the predominant planes of weakness consist of jointing or foliation parallel to the orebody and stope hanging-walls. With this predominant structure a *voussoir* arching action may develop once the immediate hanging-wall rock mass ceases to act as a continuum (Fig. 1). This arching can act along both the strike and dip of the hanging-wall. As shown in Fig. 1, deformation due to arching increases with distance from supporting abutments. Brady and Brown[2] suggested that the profile produced by the deflection of a *voussoir* beam creates an arch profile that can be described by a parabolic equation that is set to zero at the abutments and to a maximum at centre- span.

As deformation of the immediate hanging-wall rock mass increases block interlock along joints decreases and the chance of ground falls and resulting dilution increases. It can be expected that dilution, or the probability of dilution expressed as metres of slough, will increase with deformation of the hanging-wall. The present discussion, however, does not consider the influence of discrete structures that may delineate unstable blocks.

There are computer modelling techniques, such as the displacement discontinuity method, that will model continuum hanging-wall behaviour. Two difficulties arise in using these methods—the first in estimating the material properties that represent the hanging-wall rock mass and the second in identifying when the rock mass will be acting elastically, plastically or as a discontinuum. It is conceivable that all three of these mechanisms could be acting simultaneously. Even when field data are available it is difficult to calibrate the many variables that are required in a realistic computer model. The next section introduces a term that can be used to help predict hanging-wall behaviour on the basis of limited data from instrumentation.

Radius factor and effective radius factor

The previous section has discussed the factors driving deformation and dilution that tend to increase the magnitude of both with increasing distance to supporting abutments. To reflect this a term called the 'harmonic radius' has been introduced to quantify the surface geometry of a hanging-wall. The term was suggested by McGaughey[3] as a means of estimating the average distance to abutments. Distances are measured at small angular increments from points on the surface to the abutments. The harmonic average radius, R_h, or average distance to the abutments, is then given by

$$\frac{1}{R_h} = \frac{1}{n}\sum_{\theta=1}^{n}\frac{1}{r_\theta}$$

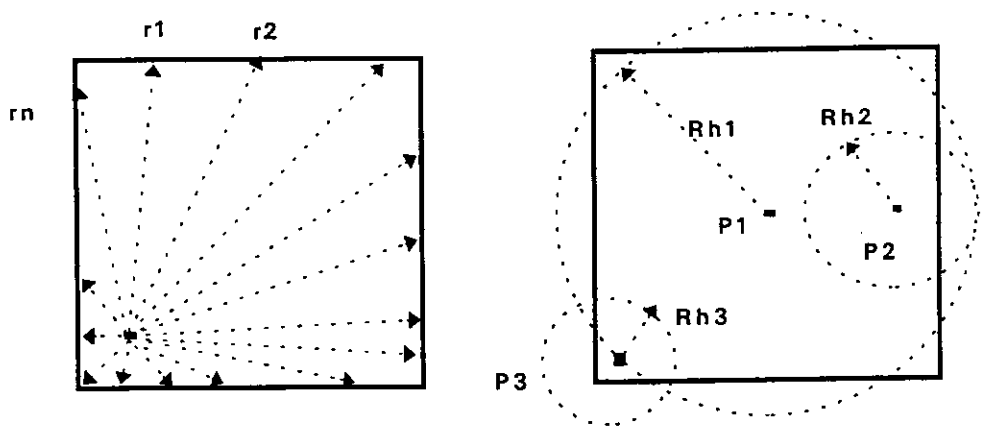
$$\therefore R_h = \frac{1}{\frac{1}{n}\sum_{\theta=1}^{n}\frac{1}{r_\theta}} \qquad (1)$$

where r_θ are the distances, measured at small angular increments, from a point at the centre of the surface to the abutments.

This form of an average is called a 'harmonic average' or 'harmonic mean'. It is used so that very large distances to supporting abutments, as would occur where hanging-wall drifts intersect an open hanging-wall, do not overly influence the calculated average.

With the introduction of the term harmonic average radius, R_h, a simple assessment of surface geometry can be made. A parameter called the 'radius factor' (RF) is proposed that is half the maximum harmonic radius of a surface. For a simple geometric surface the maximum harmonic radius and corresponding radius factor occur at the centre of the surface. For more complex geometries a program has been developed that calculates half the harmonic radius values on a grid, which allows the maximum value, corresponding to the radius factor, to be easily estimated.

The radius factor was chosen as half the maximum harmonic radius because it closely approximates the hydraulic radius of a surface. The hydraulic radius of a surface is a factor that is used in several empirical stability-design



$$\text{Harmonic average radius, } R_h = \frac{1}{\frac{1}{n}\sum_{\theta=1}^{n}\frac{1}{r_\theta}}$$

Fig. 2 Harmonic average radius, R_h, calculated at several points on a surface. (*a*) R_h is based on distances r_1, $r_2 \ldots r_n$ measured at small angular increments from point on surface to abutments. (*b*) Harmonic average radii, R_{h1}, R_{h2} and R_{h3}, for points P_1, P_2 and P_3, respectively

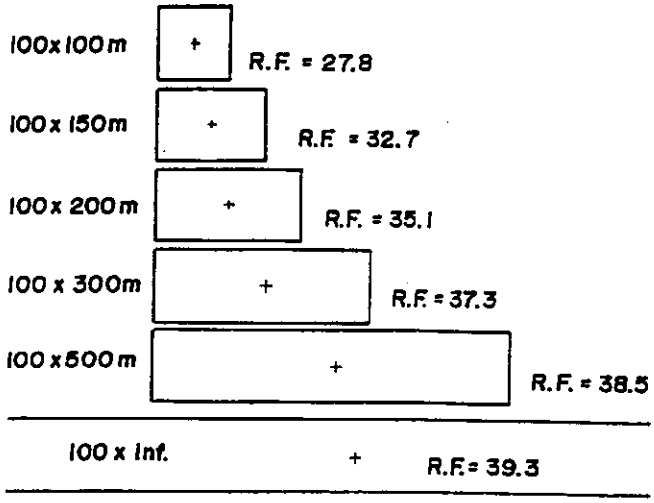


Fig. 3 Variation in radius factor (RF) for opening with constant span of 100 m and length increasing from 100 m to infinity. Radius factors calculated from centre of surface

techniques for open-stope and caving mining methods,[4–7] and it is hoped that these empirical design techniques can be related to the radius factor after further investigation.

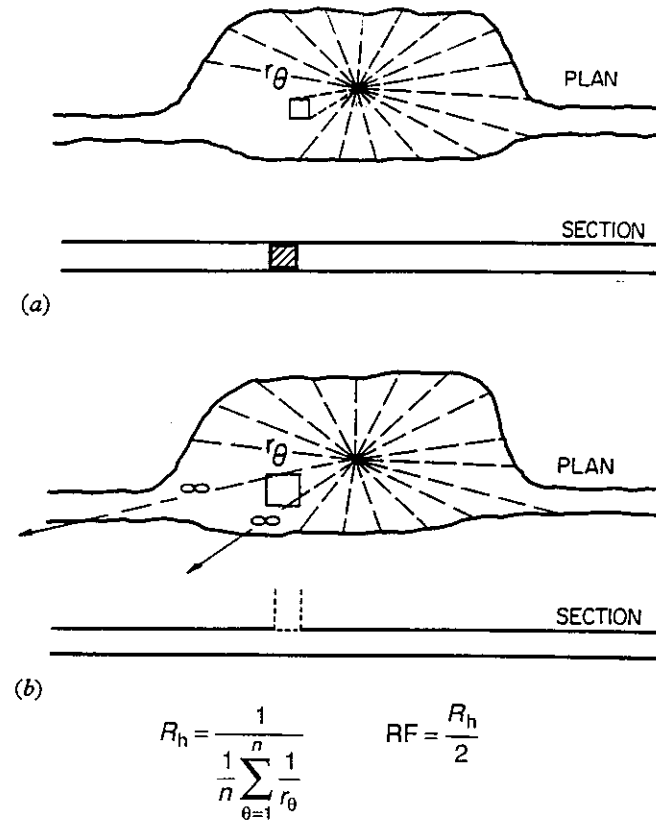
The radius factor is half the maximum harmonic radius of a surface, but a harmonic radius can be calculated for any point on a surface. Harmonic radius values, other than the maximum for a surface, correspond to a term known as the 'effective radius factor' (ERF). Fig. 2 shows a surface with harmonic average radii, R_h, calculated at several points. The R_h value is represented by circles of radius R_h, which are at a maximum at the centre of a surface and decrease near the abutments.

Radius factor and geometry

The radius factor of a hanging-wall surface is influenced by the distance to each abutment. In the case of a rectangular surface opening both the span and the strike length influence the RF value; however, when the long dimension is three times the short dimension increasing the long dimension further can only increase the RF value by about 5%. Fig. 3 shows the variation in RF values as an opening 100 m square increases in extent in one direction until it becomes a 100 m wide tunnel.

One advantage of the radius factor and effective radius factor values is the ability to assess complex geometries. Fig. 4 shows plan views of two geometries: the first case (Fig. 4(*a*)) is that of a small pillar located near the middle of a large back and in the other a raise has been driven near one wall of a back (Fig. 4(*b*)). The presence of the pillar results in smaller harmonic average radius values for the surface and a smaller RF value. In the case of the raise no support is given by the abutment that is shadowed by the raise. To take the raise into account radius vectors that intersect the raise are treated as infinitely long for the calculation of harmonic average radius. As shown in Fig. 4, the radius factor can be used to assess these geometries. The influence of both the pillar and the raise increases near the centre of the opening and reduces near the abutments, as would be expected in the field.

A program has been written at the Noranda Technology Centre, Montreal, that automatically calculates effective radius factor values for any point on a surface or grid pattern. The program recognizes the presence of pillars, raises or brows and calculates the ERF values using any angular increment desired for determining the average harmonic radii.



$$R_h = \frac{1}{\frac{1}{n}\sum_{\theta=1}^{n}\frac{1}{r_\theta}} \qquad RF = \frac{R_h}{2}$$

Fig. 4 Influence of (*a*) pillars and (*b*) raises on calculation of radius factor

Case histories

Detour Lake mine—design application

Placer Dome Inc.'s Detour Lake mine is a 2800 t/day gold operation north of Timmins, Ontario, Canada. Mining is by mechanized cut-and-fill and longhole stoping.[8] The orebody is near-vertical and has a width of approximately 6 m in the study area. In 1992 the mine entered into a research agreement with the Canada Centre for Mineral and Energy Technology (CANMET) to study dilution in one of its sub-level retreat stopes.

Fig. 5 is a longitudinal view of the west sub-level retreat stope showing the frequent and irregularly spaced pillars that are left for hanging-wall support. Most of the stope was mined open with the addition of fill only near the end of its life for long-term regional support. Owing to the sub-level retreat mining method dilution was only a concern when it occurred near the retreating mining front. Since the throw of the muck is about 15 m, any hanging-wall dilution that occurred more than 15 m away from the trailing mining front would be unlikely to be mucked.

During the course of mining cavity-monitoring system (CMS) laser surveys were conducted to determine the dimensions of the stope and the degree of hanging-wall dilution.[9] Although only limited data were obtained from the surveys because of the narrowness of the stope and restricted access, useful trends were obtained. The effective radius factor term proved to be a useful method of evaluating the stability of the zone of open hanging-wall near the retreating mining front. The CMS surveys indicated that little hanging-wall sloughing occurred along supported sub-level horizons. It was found, however, that sloughing of up to about 2 m occurred when the effective radius factor values between sub-levels exceeded 12. One metre of slough is equivalent to about 15% dilution. Fig. 5 shows the area of the hanging-

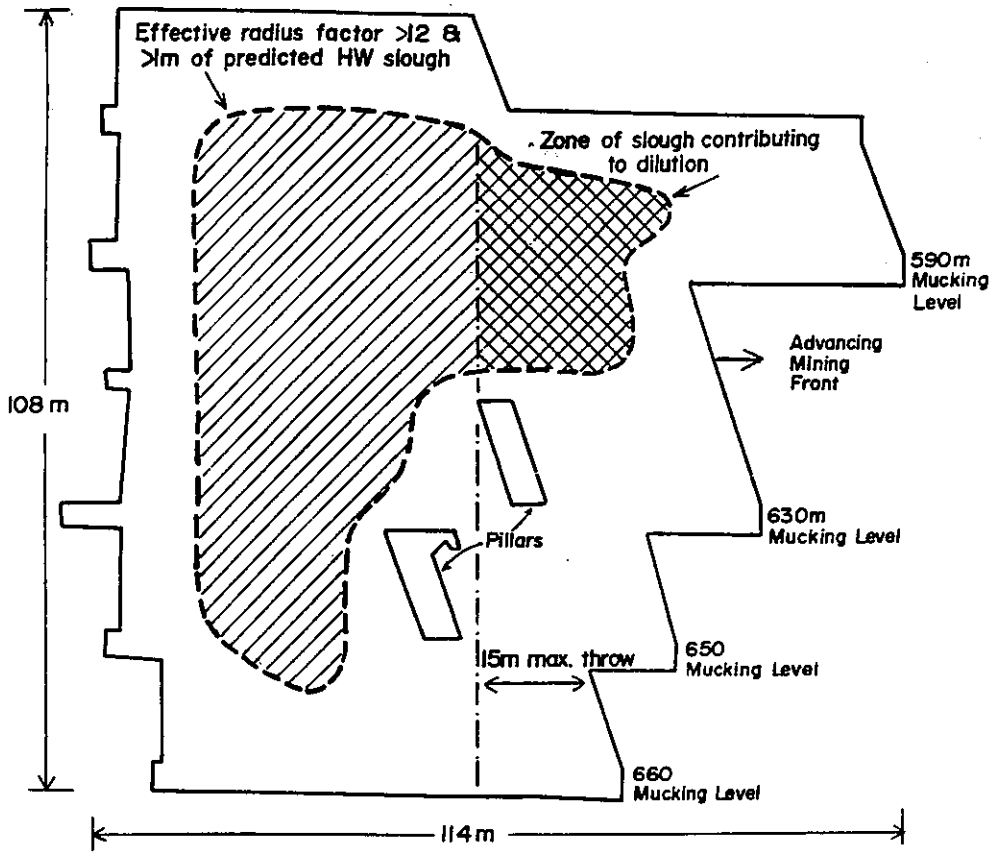


Fig. 5 Longitudinal view, looking north, of west sub-level retreat stope at Detour Lake mine. Area where effective radius factor values exceed 12.0 and more than 1 m of hanging-wall (HW) slough is expected is indicated. Where this zone of expected slough could contribute to mined dilution is also shown. Orebody is near-vertical in area shown

wall where the effective radius factor values exceeded 12 and the resulting slough could be mucked with the ore. Understanding which areas of the hanging-wall produce significant dilution enables the mine to adjust the mining retreat front and relocate pillars to minimize the mucking of this dilution with ore.

Brunswick mine—prediction of surface deformation

Brunswick mine is located near Bathurst in New Brunswick, Canada. The mine produces 10 500 t/day lead–zinc–copper–silver ore. A geomechanics research project, sponsored by Brunswick Mining, the Noranda Technology Centre and the Mining Research Directorate, was carried out at the mine in 1992–93. The study was conducted to examine the geomechanics problems associated with mining in the highly stressed conditions that are found in the sill pillars.[10] In the area of the study the orebody is about 5–10 m wide and dips approximately 70–75° to the west. Stresses in the sill pillar area were estimated from modelling to be about 70 MPa and the intact unconfined compressive strength of the ore is about 220 MPa.

As part of the study, extensometers were installed to monitor hanging-wall behaviour with mining. One of these instruments, extensometer no. 2, was undercut at an early stage and it was possible to follow the increasing deformation of the hanging-wall as mining continued. Fig. 6 shows several longitudinal views of the hanging-wall, each representing a mining step. The location of the bottom extensometer anchor is shown and an effective radius factor value has been calculated at each mining step, corresponding to the extensometer anchor location. The final stope was mined to 58 m on strike and about 45 m on dip. Cable support was installed from the sub-level, but extensive dilution accompanied the final stope blast.

Graphs of the deformation between the collar and the extensometer anchor closest to the hanging-wall and of the strain between the anchor closest to the hanging-wall and the next nearest anchor 2 m above are presented as Fig. 7. Both graphs show a very consistent relationship between effective radius factor and hanging-wall deformation. It should be noted that the strain measured between the bottom two anchors (Fig. 7(*b*)) reaches a maximum value of about 1.5%, well in excess of elastic strain values. Furthermore, the strain level at which cable support yields (also indicated in (Fig. 7(*b*)) has been greatly exceeded.

The final mining step, blast 19, resulted in extensive hanging-wall failure over much of the stope. A maximum effective radius factor value at the centre of the hanging-wall of 12.4 was calculated. If the strain versus effective radius factor values based on the trends shown in Fig. 7 can be extrapolated, a maximum strain value for the immediate hanging-wall of 5% would be projected from the available data. It is evident that strain levels of this magnitude could not be supported by the hanging-wall rock.

Discussion

Both hanging-wall dilution and deformation are caused by a complex response of the rock mass to stress redistribution and arching. If the case of discrete structures that may delineate unstable blocks is ignored, both deformation and dilution can be expected to increase with distance from supporting abutments. Two new parameters, radius factor (RF) and effective radius factor (ERF), have been intro-

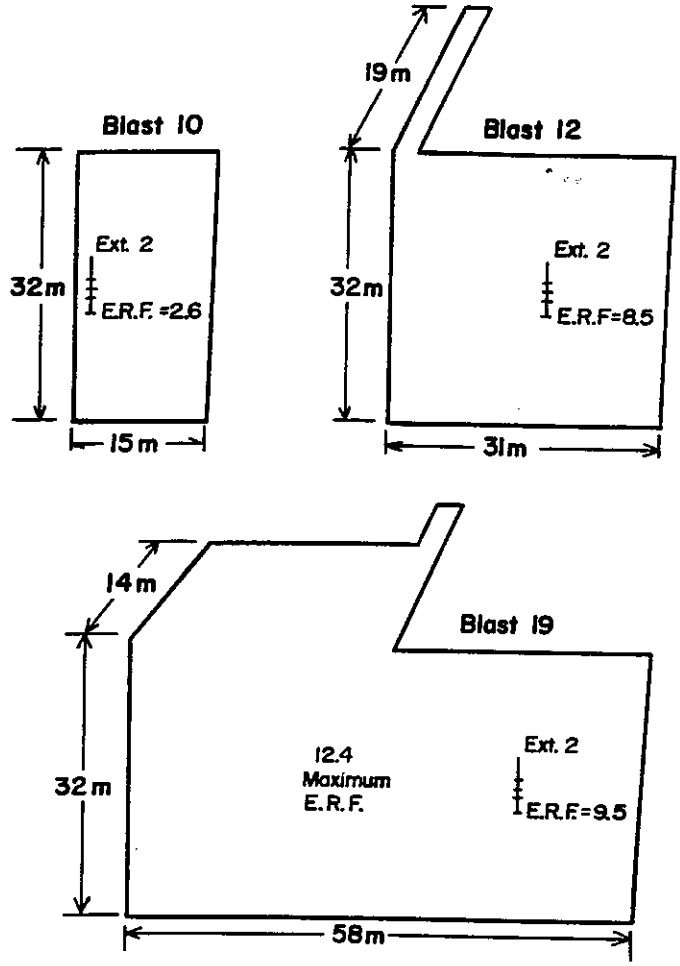


Fig. 6 Brunswick mine 16N stope looking west showing longitudinal views of three stages of mining (blasts 10, 12 and 19) and corresponding effective radius factors (ERF) for hanging-wall extensometer (Ext. 2). Orebody dips 70–75° to west in this area

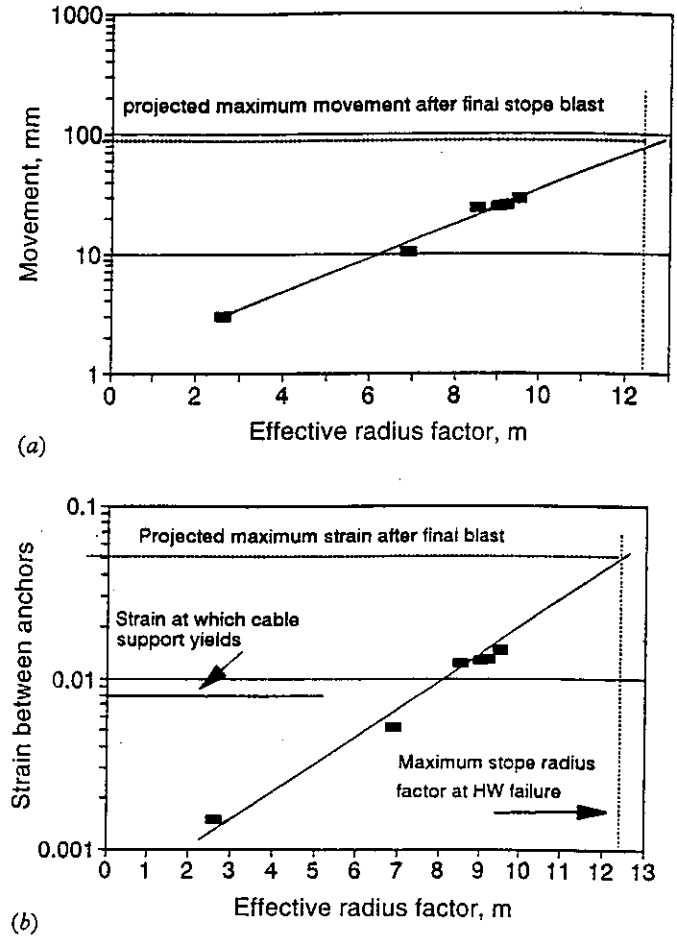


Fig. 7 Graphs showing deformation of and strain in immediate stope hanging-wall with increasing effective radius factor, Brunswick mine, stope 16N, extensometer no. 2. (*a*) Movement between collar and bottom anchor versus effective radius factor; (*b*) strain between bottom anchors versus effective radius factor

duced. Both factors are based on the calculation of an average distance to supporting abutments. The effective radius factor can be determined at any point on a surface and has been related to the corresponding deformation at that point. It has also been possible to relate strain and deformation of the immediate hanging-wall to the ERF value. The RF value is the maximum ERF value for a surface and, for simple geometries, is calculated at the centre of the surface.

A trend linking ERF values and deformation can be determined with quite limited data from instrumentation. It is then possible to predict the response of the hanging-wall to further mining by extrapolation and also to project hanging-wall deformation beyond the instrumented area. With this information it may be possible to design the stiffness of support to match the expected response of the hanging-wall to mining. With more case histories of hanging-wall deformation it may be possible to start relating the change in slope of the curve of ERF versus deformation to the mode of deformation and the approach of instability.

A relationship was also found between the ERF values on the surface of a hanging-wall and the resulting depth of slough or dilution. As the distance to supporting abutments increased the depth of slough and the ERF value increased. The example of Detour Lake mine showed that less than 1 m of slough could be expected when the ERF values were less than 12. With this relationship the mining front and the placement of pillars could be planned better. Cavity-monitoring laser surveys are becoming more common in underground mines. The ERF factor should assist in the interpretation of the new data on mining dilution that will be provided by these surveys.

Acknowledgement

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